Flowchart Practice Exercises

Fundamentals of Programming I · Chapter 1

For every exercise:

- (a) Write the analysis: input, process and output.
- (b) Write the pseudocode.
- (c) Draw the flowchart. Use an oval for Start/End, a parallelogram for input/output, a rectangle for a calculation and a diamond for a decision.
- (d) Trace it by hand with the test data.

Try each one on paper before looking at the solutions page.

Exercise 1: Celsius to Fahrenheit

Easy · sequence

Read a temperature in degrees Celsius and display it in degrees Fahrenheit. Formula: F = C × 9 / 5 + 32.

Test data: C = 100 gives 212. C = 0 gives 32.

Exercise 2: Voting eligibility

Easy · decision

Read a person's age and display whether they are eligible to vote (age 18 or more).

Test data: Age 20: eligible. Age 17: not eligible. Age 18: eligible.

Exercise 3: Positive, negative or zero

Easy · nested decision

Read a number and display whether it is positive, negative or zero.

Test data: 5: positive. -3: negative. 0: zero.

Exercise 4: Largest of three numbers Medium · decisions

Read three numbers and display the largest.

Test data: (4, 9, 6) gives 9. (7, 7, 3) gives 7. (2, 2, 2) gives 2.

Exercise 5: Multiplication table Medium · loop

Read a number n and display n × 1, n × 2, … up to n × 10.

Test data: n = 3 prints 3, 6, 9, … 30.

Exercise 6: Factorial Medium · loop

Read a number n and display n! = 1 × 2 × … × n.

Test data: n = 5 gives 120. n = 0 gives 1.

Exercise 7: Average of marks until -1 Harder · loop with a stop value

Read marks one by one until the user enters -1 (which is not a mark). Display the average of the marks. If no marks were entered, display a message.

Test data: 60, 80, 70, -1 gives 70. Only -1 gives “no marks entered”.

Exercise 8: Largest of N numbers Harder · loop with a decision

Read N, then read N numbers and display the largest.

Test data: N = 4 with 12, 7, 19, 5 gives 19.